

Cell Physiology

Interesting facts about cell and cell membrane:

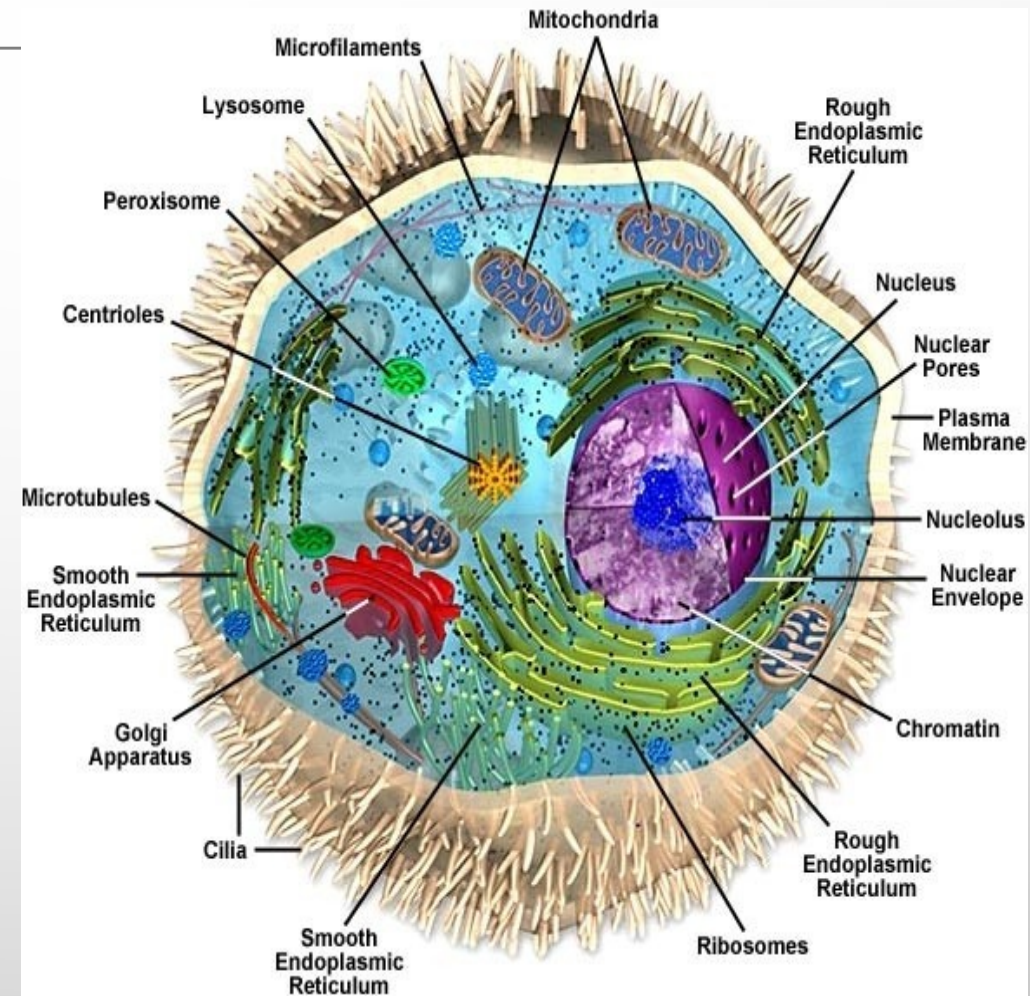
- ❑ Scientists estimate that our bodies contain anywhere from 75 to 100 trillion cells, all work in perfect harmony.
- ❑ A typical cell is **10 square micrometers** in size and **1 nanogram in mass**!
- ❑ The cell membrane is ridiculously small. It's so small, in fact, that you'd have to stack more than 10,000 cell membranes on top of each other to equal the thickness of a standard piece of the paper. Yet despite this small thickness, the total surface area of cell membrane in the average human body is incredibly large. There's more than **27871** square meters of it in the human body, which is enough to fully cover 4 American football fields.
- ❑ The amount of information that the cell membrane protects is incredible. It's essentially like having your own HDD operating within every cell. What the cell membrane does, however, is protects these data and actually use this data library. It directs the information to go to different places in the body and it helps to keep people functional and healthy.
- ❑ The of phospholipids of the membrane acts a lot like the insulation of a double- glazed window while helps to facilitate movement of specific selective substances and resisting molecules that shouldn't be entering the cells

AFTER THE END OF THESE LECTURES, YOU SHOULD BE ABLE TO . . .

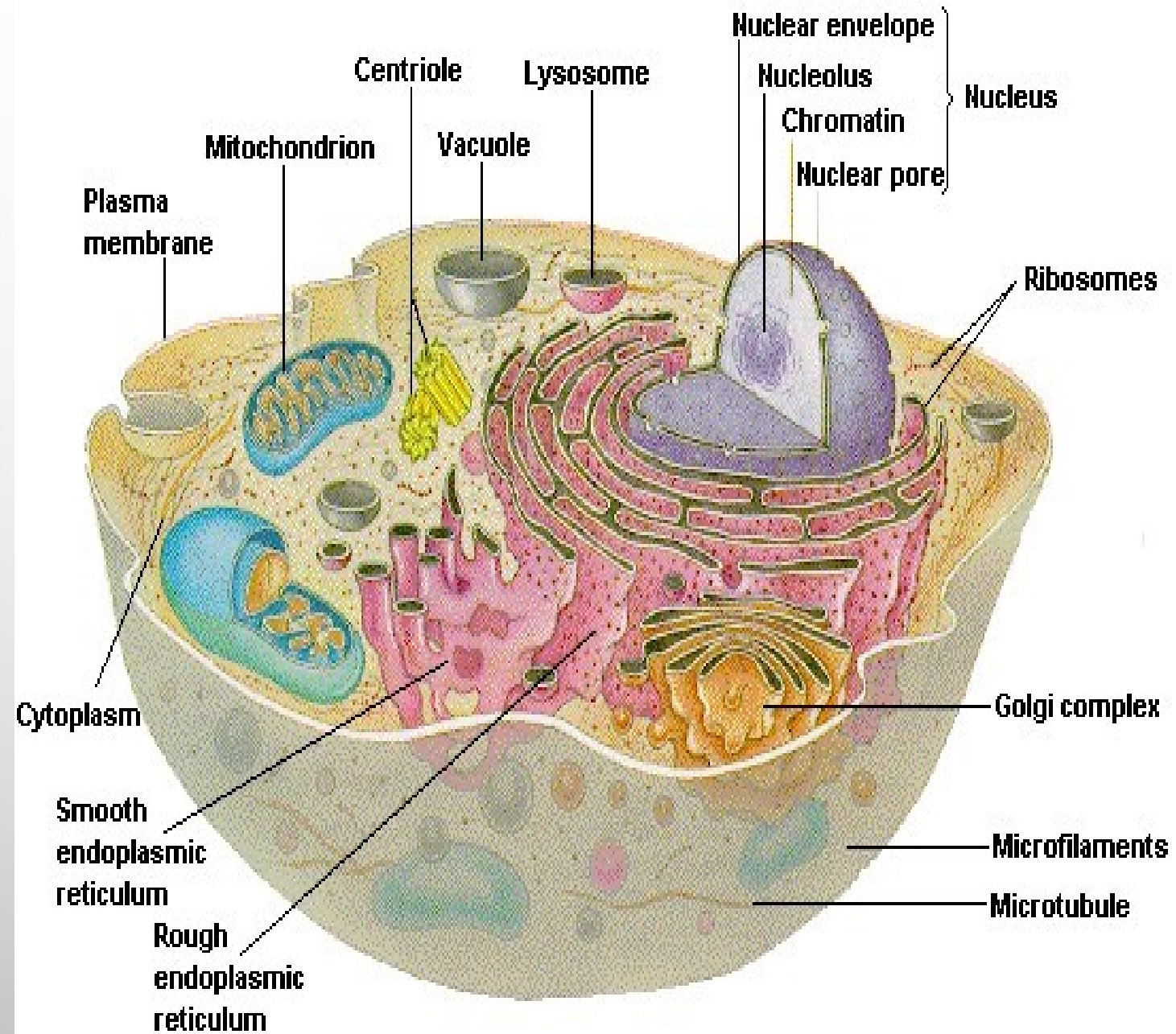
1. Describe the structure and functions of the plasma membrane, glycocalyx, and ion channels.
2. Expound the functions of cilia and flagella.
3. Explain the means of cell-to-cell adhesion.
4. Describe how the cells communicate with each other (signal transduction or cell signaling).
5. Expound the ways by which cell regulates the intracellular $[Ca^{2+}]$ and $[H^+]$.

CELL PHYSIOLOGY

CELLS ARE THE BASIC UNIT OF LIFE



Cell structure



CELL STRUCTURE

Water □ 70-85%

Proteins □ 10-20%

Lipids □ 2% (mainly phospholipids & cholesterol)

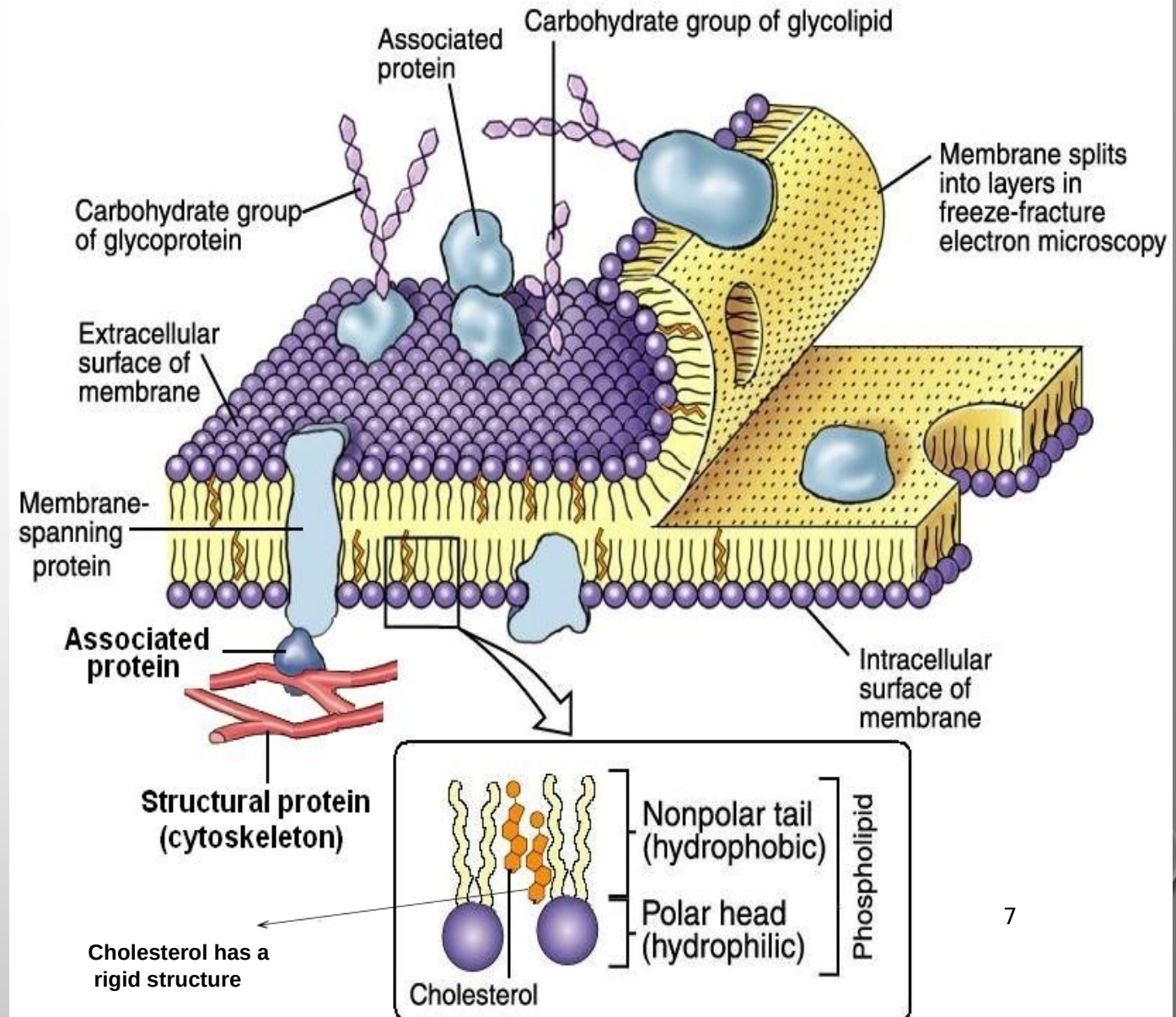
Carbohydrates □ 1%

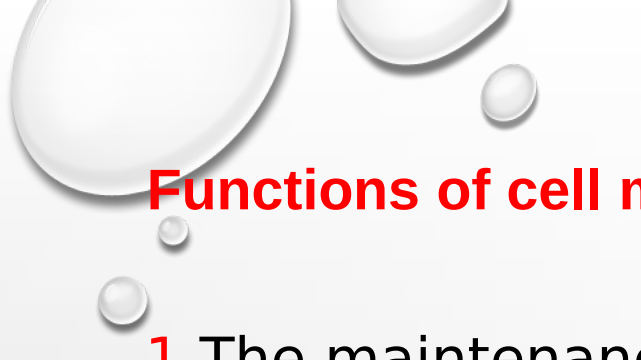
Electrolytes □ potassium, magnesium, phosphate, sulfate, bicarbonate, and small quantities of sodium, chloride, and calcium.

Cell membrane (plasma membrane): The membrane made up **lipids (40% of the membrane)** and **protein (60% of the membrane)**.

lipid-soluble substances: O₂, CO₂, N₂, anesthetic gases, steroid hormones, alcohol)

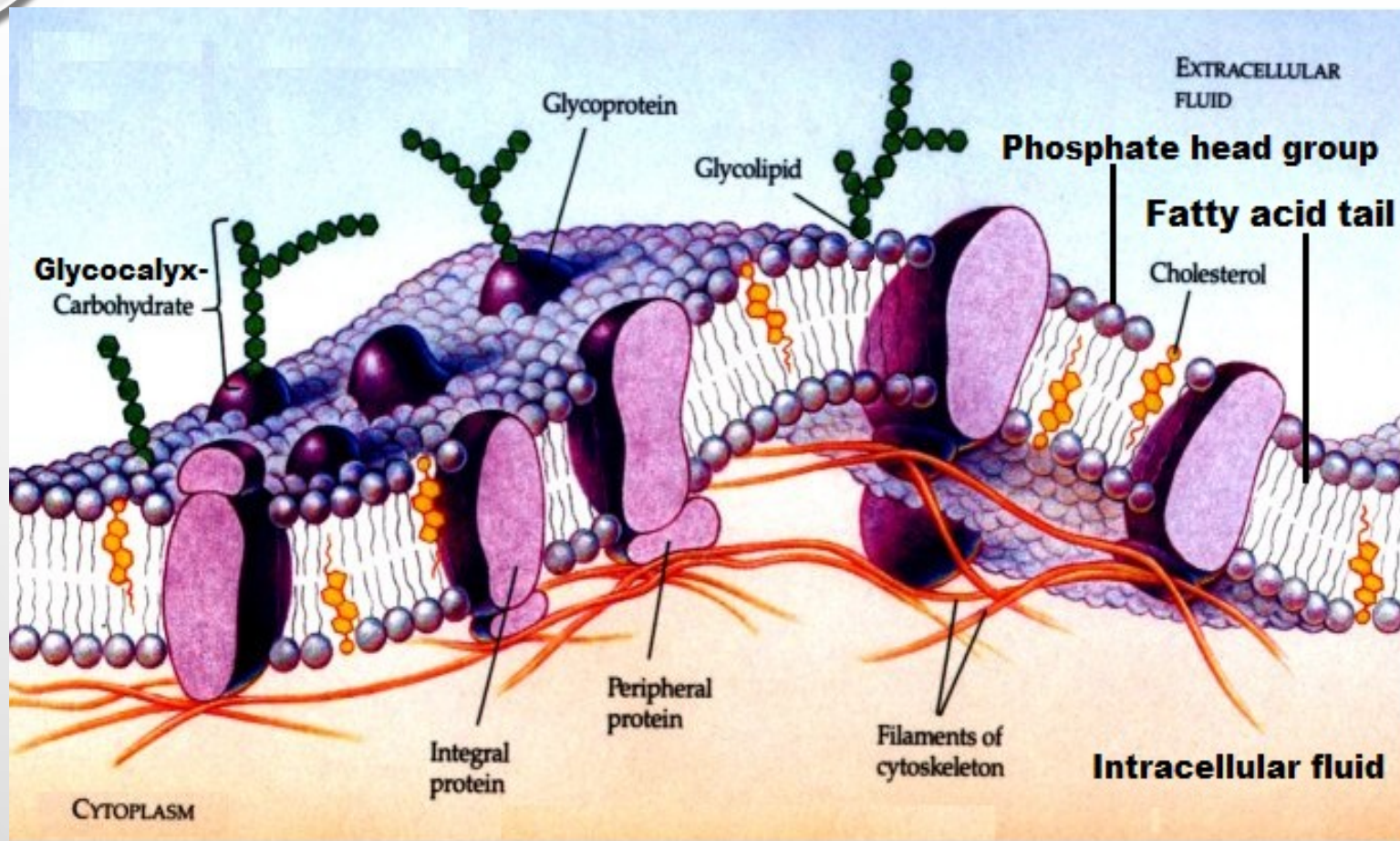
water-soluble substances: Electrolytes, glucose, urea, H₂O)





Functions of cell membrane:

1. The maintenance of cell shape and structure.
2. A transport function: by selective permeability can maintain ionic composition, osmotic pr., pH, and contents of the cell.
3. Intercellular communication: involving signal transduction that mediate nerve transmission, hormone release, muscle contraction.
4. Intercellular adhesion by specialized junctions.
5. Directed cell movement.

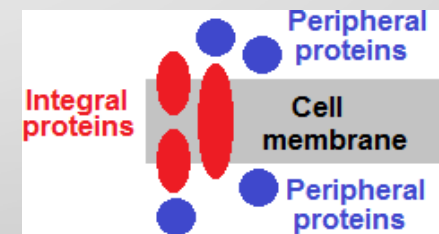


Integral prot.:

Ion channels proteins
 Pumps
 Transport proteins (carriers)
 Receptors
 Cell adhesion molecules
 Antigens & recognition proteins
 Enzymes

Peripheral prot.

Cytoskeletal
 Regulatory proteins
 Enzymes



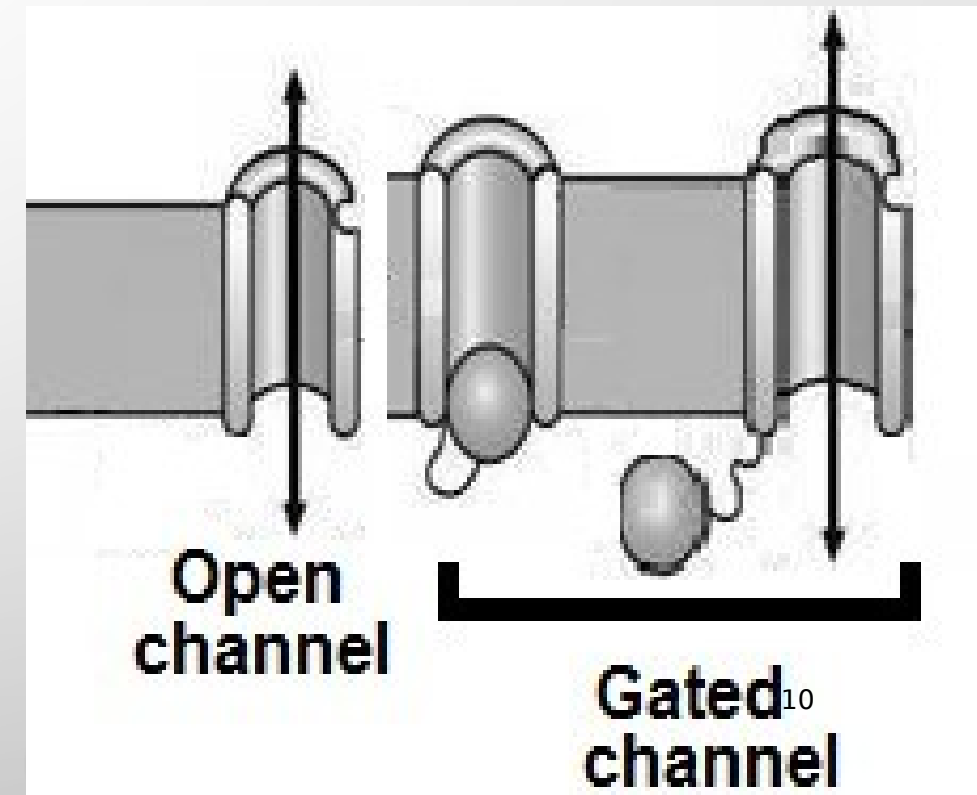
The general characteristics of transmembrane ion channel proteins:

- 1- **Specificity:** highly selective for transport of one or more ions or molecules.
- 2- They are either open channels or gated channels (i.e. can be opened or closed by gates)

(A) Voltage gating (voltage-gated channel), Na and K channels

(B) Chemical or ligand gating (chemical or ligand-gated channel), Ca-sensitive K channel

(C) Physical gating (physical-gated channel), in membrane of pressure receptors

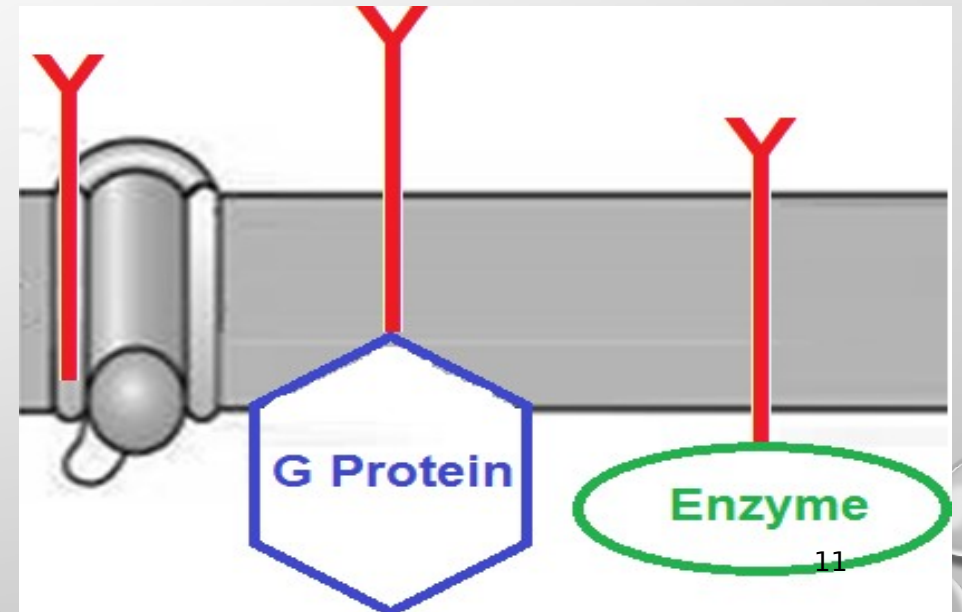


The receptors: They are proteins or glycoproteins. They are located on the surface of cell, or within the cytoplasm or nucleus. Receptors have the properties of:

- ⊙ Specificity
- ⊙ High affinity
- ⊙ Limited capacity
- ⊙ Down & up-regulation

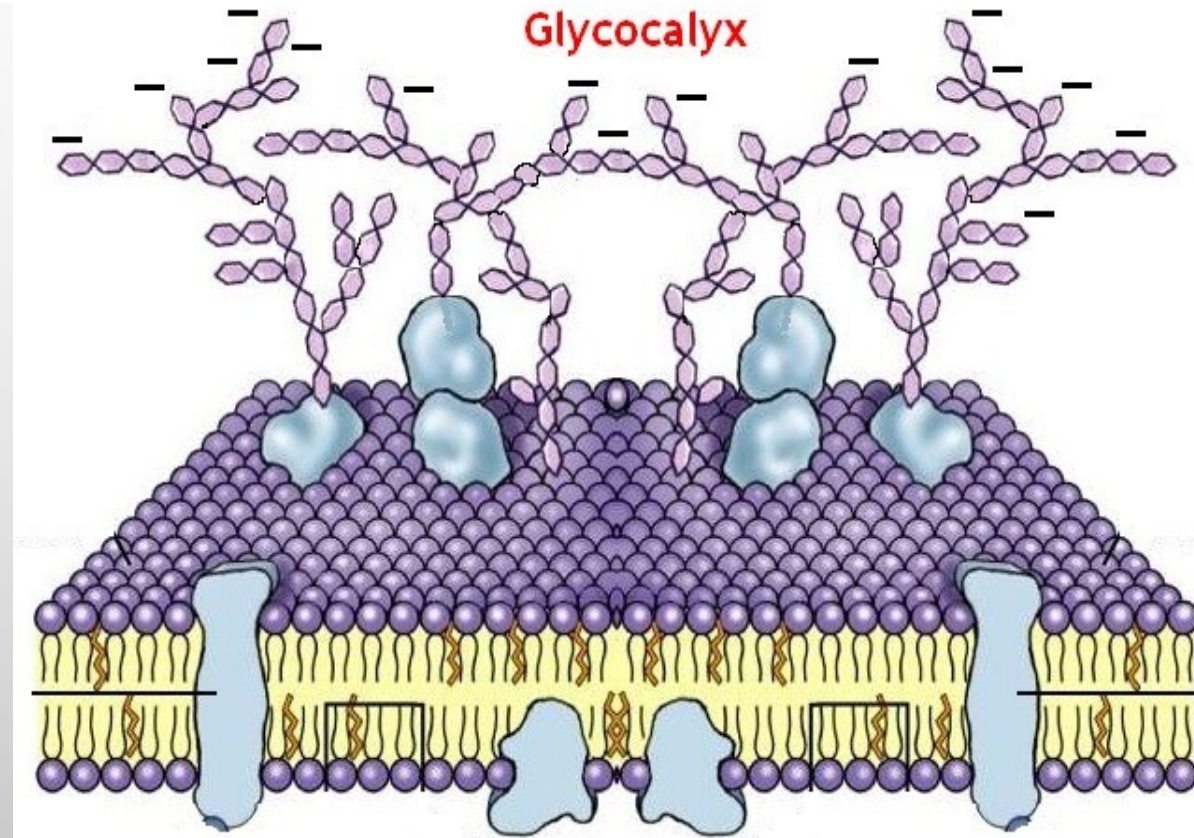
Classification of cell membrane receptors:

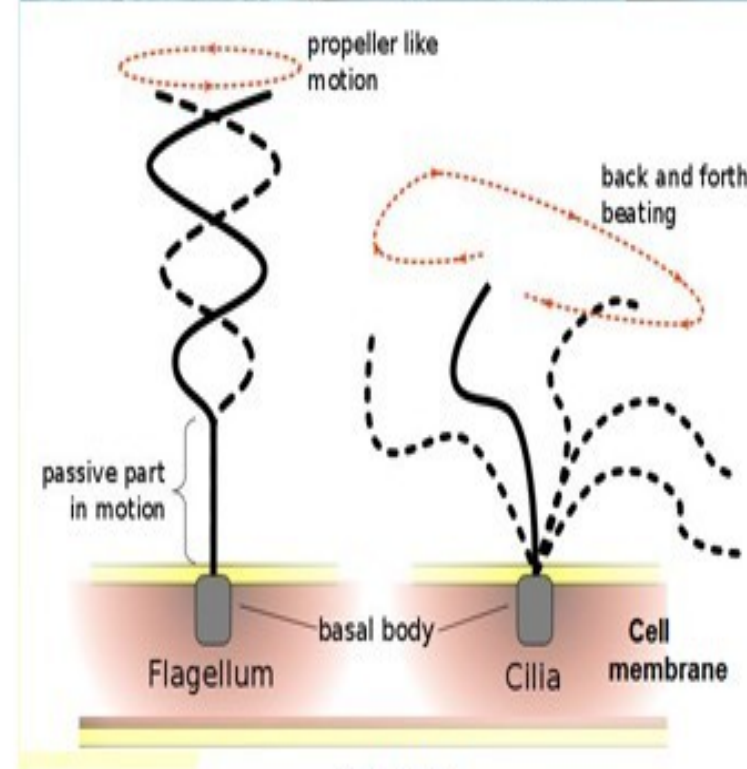
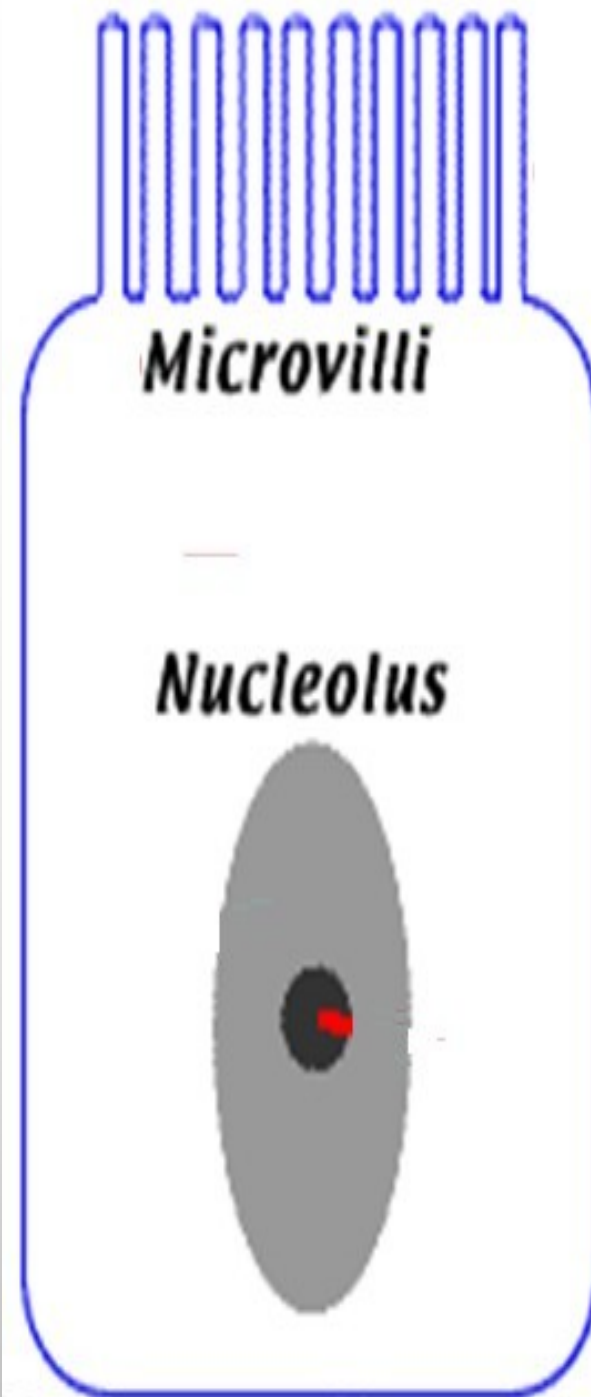
1. Gated channel-linked receptors
2. Catalysis-linked receptors
3. G-protein-linked receptors



GLYCOCALYX

- A. Repels other negative objects.
- B. Attaching the cells to each other.
- C. Receptor
- D. Enters into immune reactions □ antigens markers that identify the cells of an individual as “self.”





Cilia

CILIA

Are non-motile (more widespread) or motile (less widespread) thread-like projections through the cell membrane; each is anchored by a basal body just within the membrane

In inner ear □ play a role in the sense of balance **non-motile**

In the retina □ form the light absorbing part of the receptor cells **non-motile**

In kidney tubules □ monitor fluid flow through the kidney tubules **non-motile**

In the nose □ bind odor molecules **non-motile**

In the respiratory tract and the uterine (fallopian) tubes □ sweeping **motile**

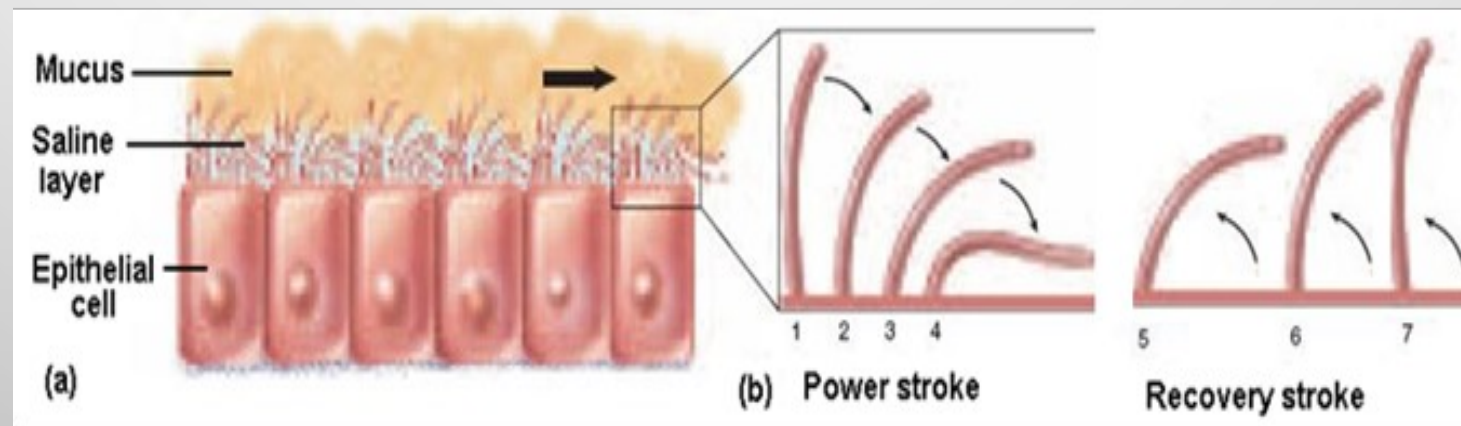


Figure 1.5: Cilia and its movement.

Cell-to-Cell Adhesions:

A. Cell adhesion molecules (CAMs): They are integral membrane proteins that have cytoplasmic, transmembrane and extracellular domains.

Immunoglobulin-like adhesion molecules, Integrins, Cadherins, Selectins.

B. The extracellular matrix (ECM, biological “glue”): Gel (ground substance) + network of fibrous proteins

Collagen □ flexible non-elastic fibers

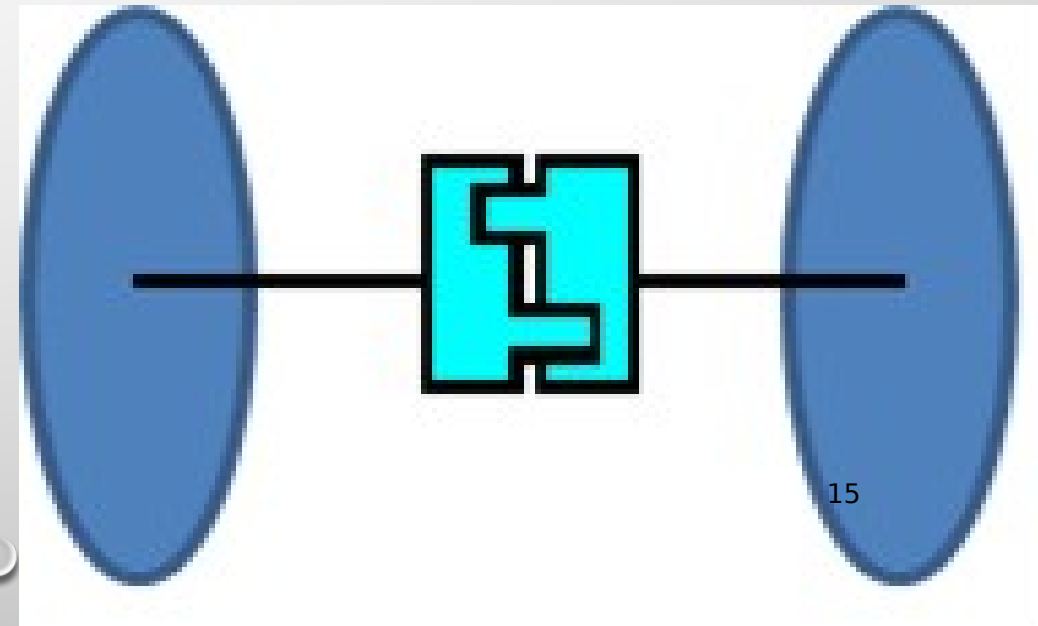
Elastin □ flexible elastic fibers

C. Specialized membrane junctions:

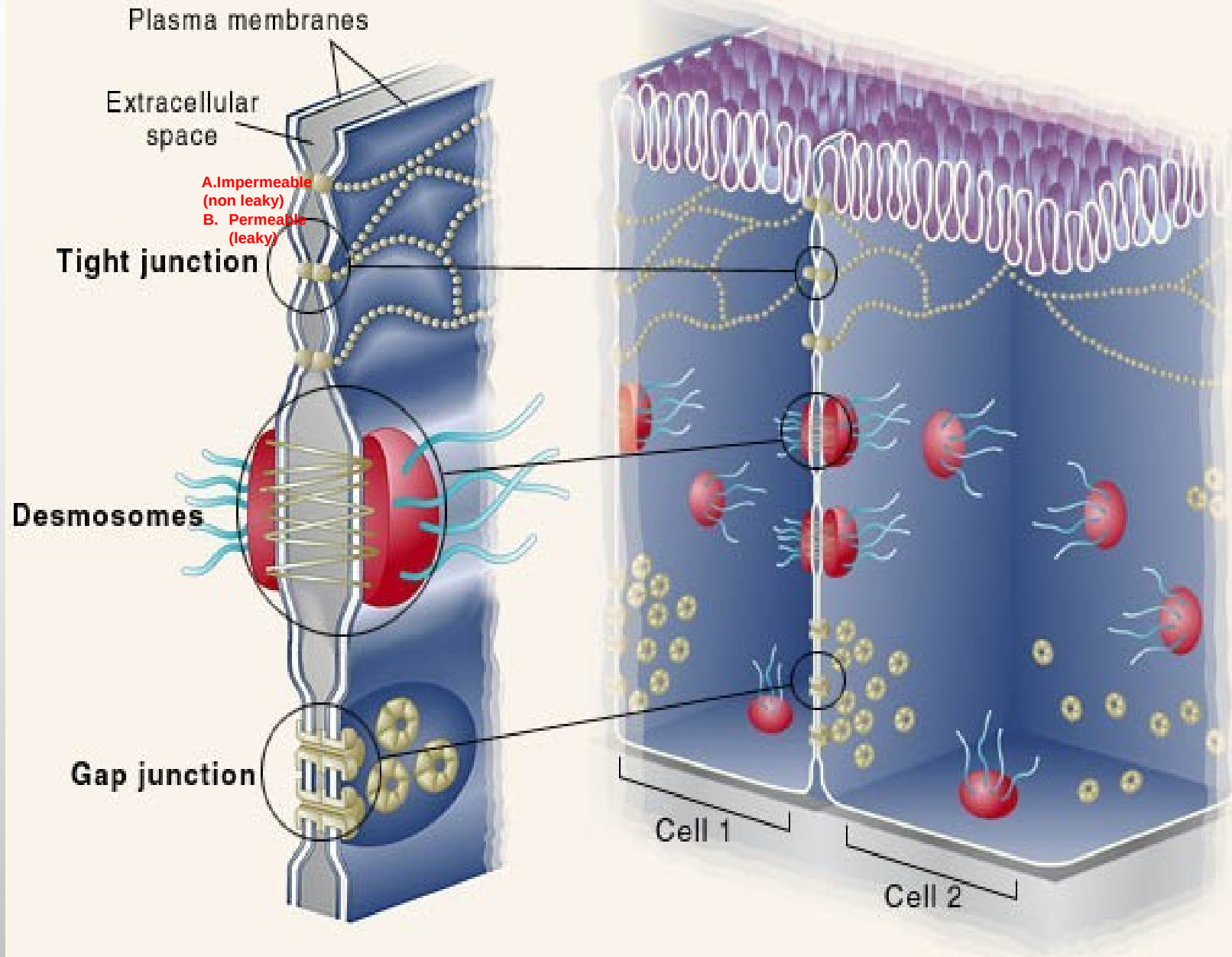
Tight junction

Gap junctions

Adheren or Desmosome junctions



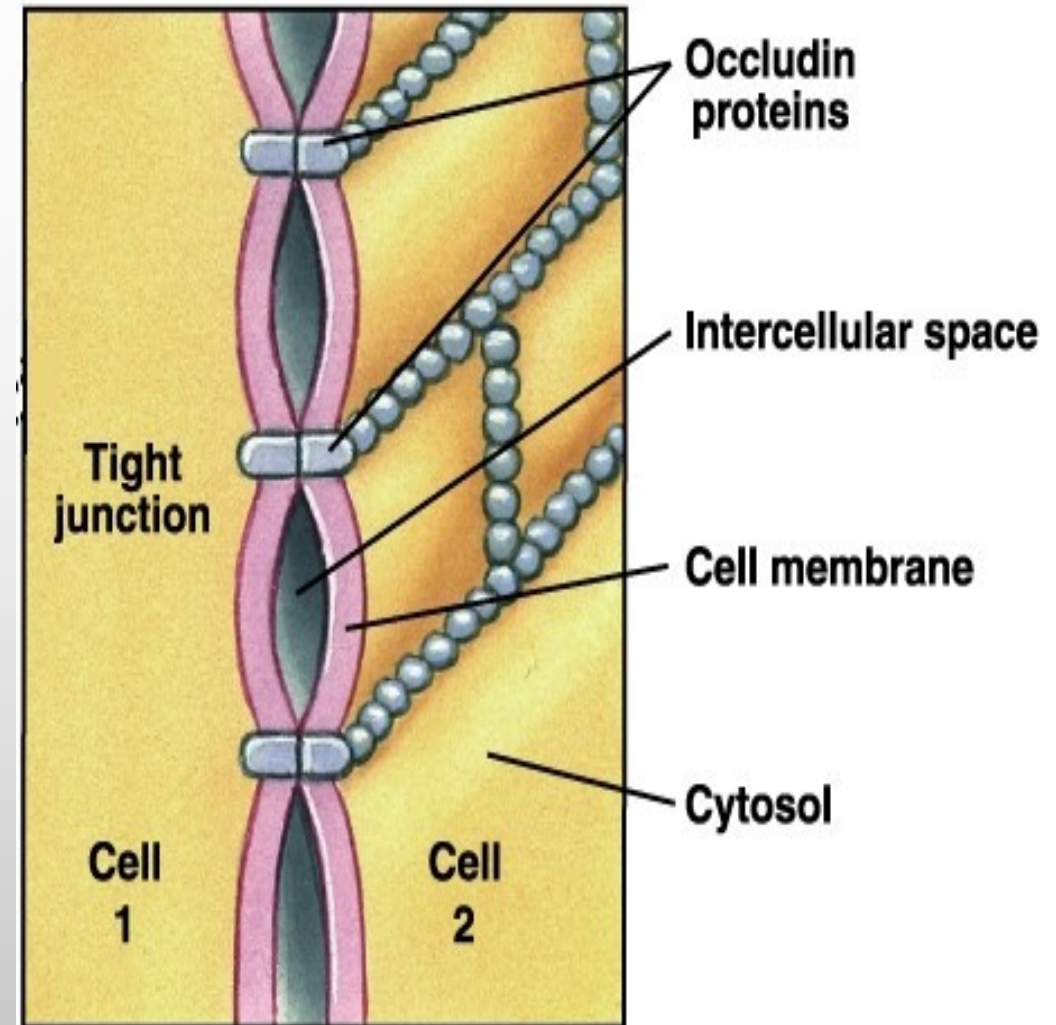
Specialized cell junctions



MEMBRANE JUNCTIONS: TIGHT JUNCTION

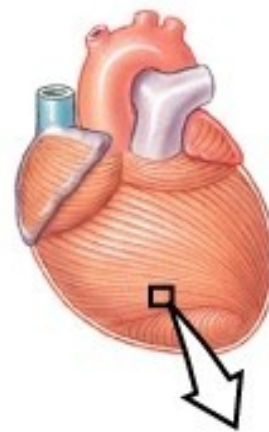
- A. Impermeable (non leaky),** separate two compartments having different chemical compositions
- B. Permeable (leaky),** allows substances to pass depending on the size, charge, and characteristics of tight junction

Tight junction

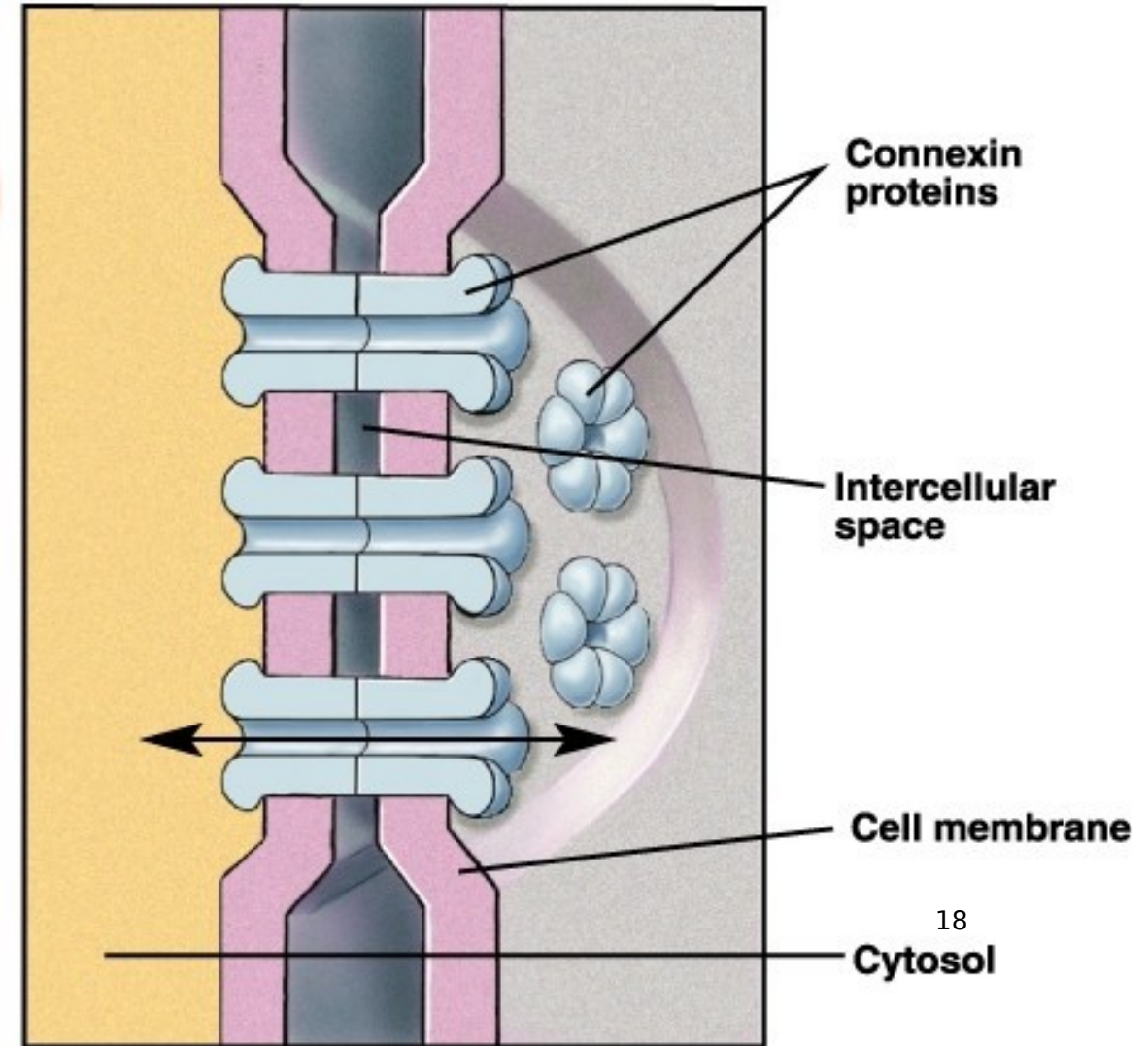


Membrane junctions: Gap junction

- There is a channel between two adjacent cells
- Cytoplasm of two cells are connected
- Transfer of ions without entering ECF
- Cardiac and smooth muscle cells

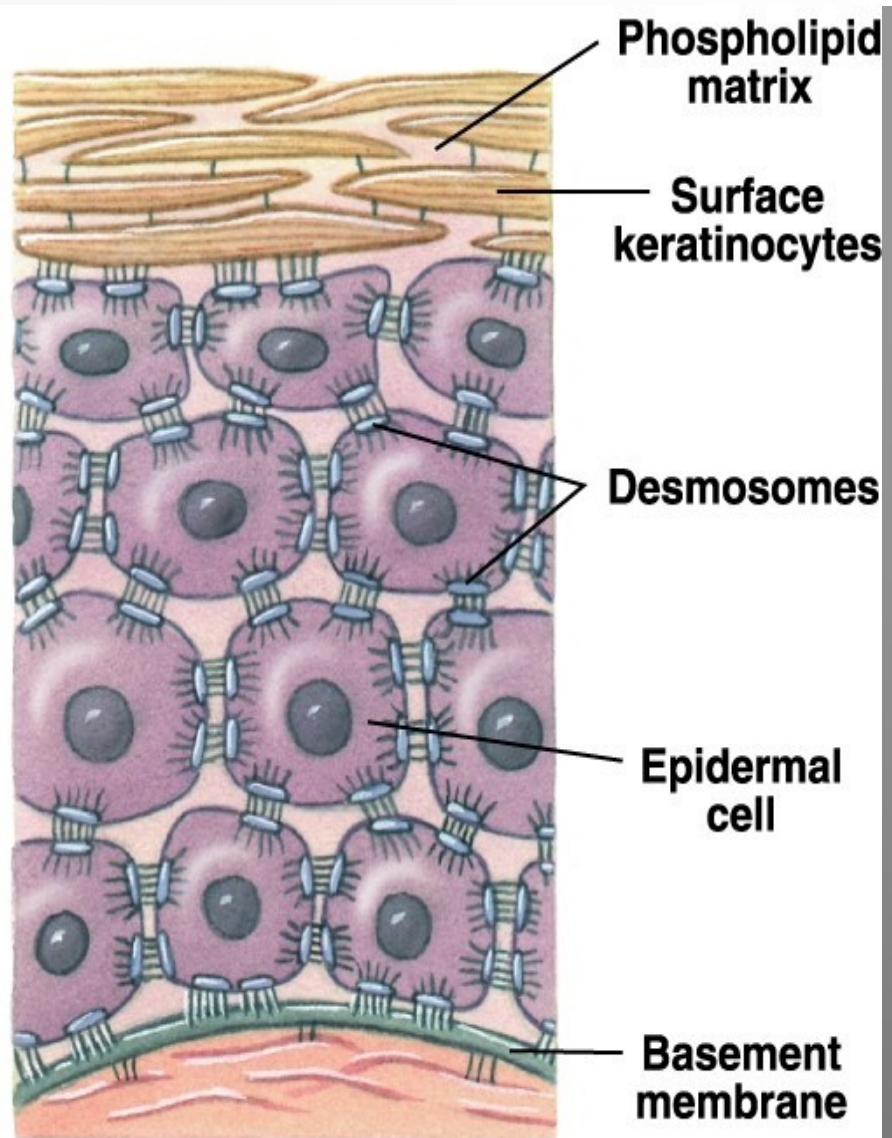


Gap junction

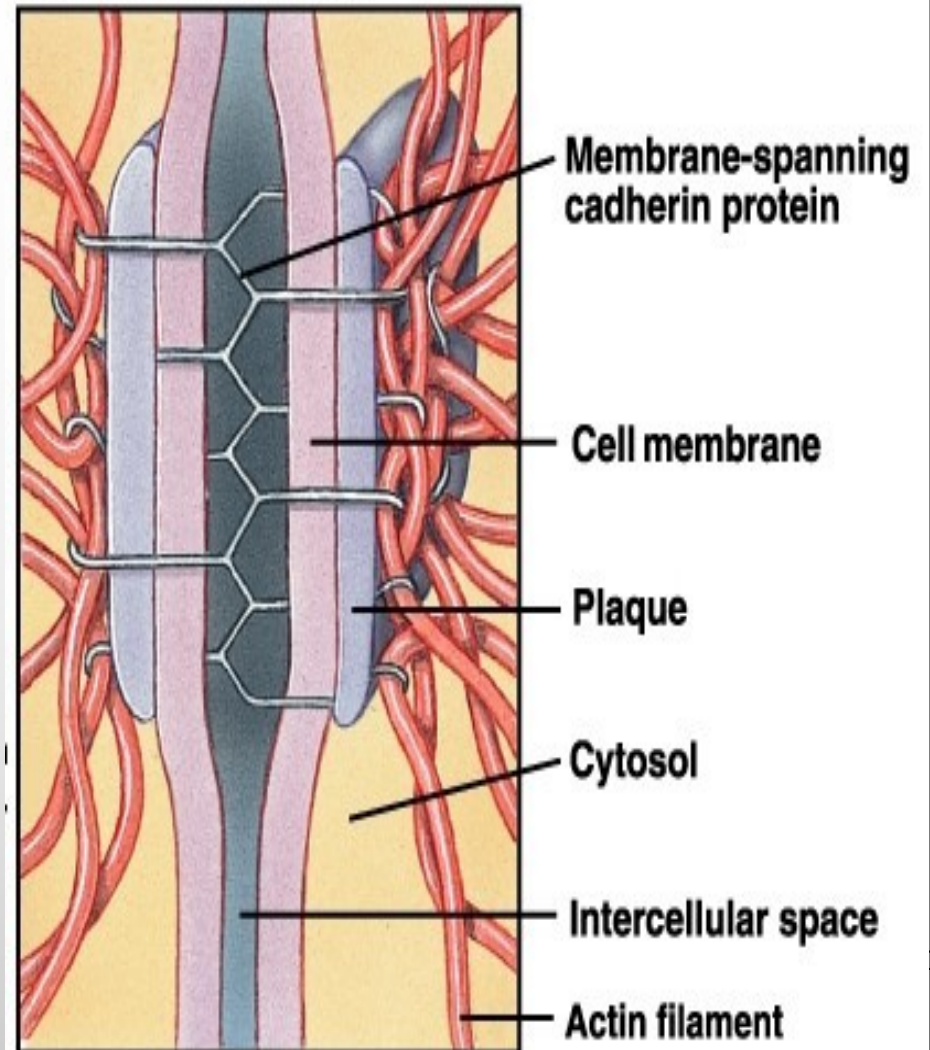


MEMBRANE JUNCTIONS: DESMOSOME JUNCTION

Skin & heart muscle



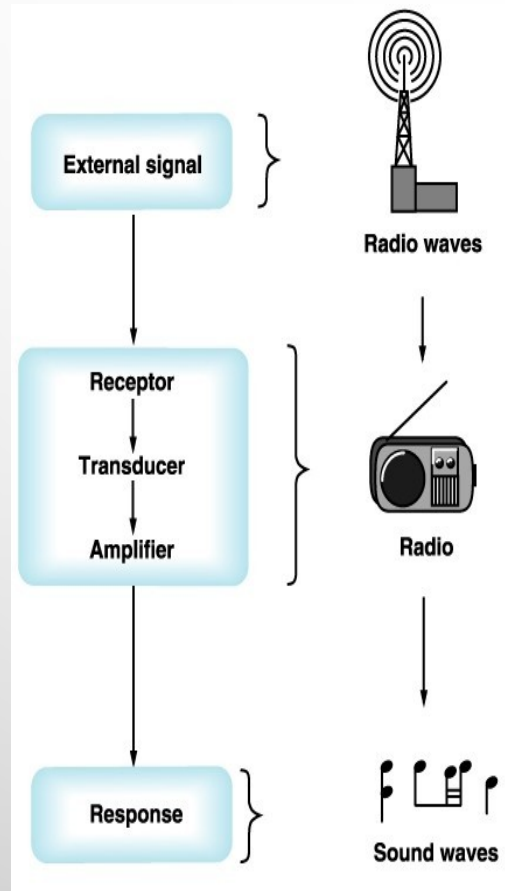
Adherens junction



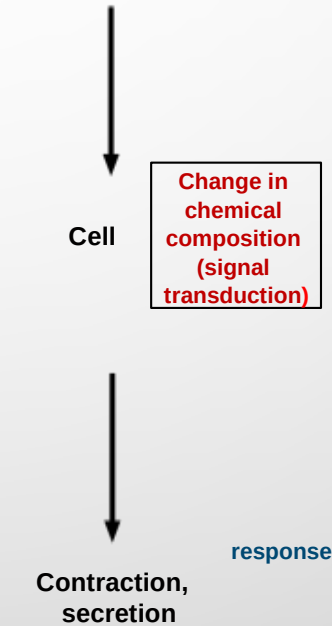
Intercellular communication (signal transduction)

change in metabolic activity, contraction or relaxation, secretion, release of another signaling substance, cell growth and division, and cell death (apoptosis).

The process by which a cell converts an input signal into a response

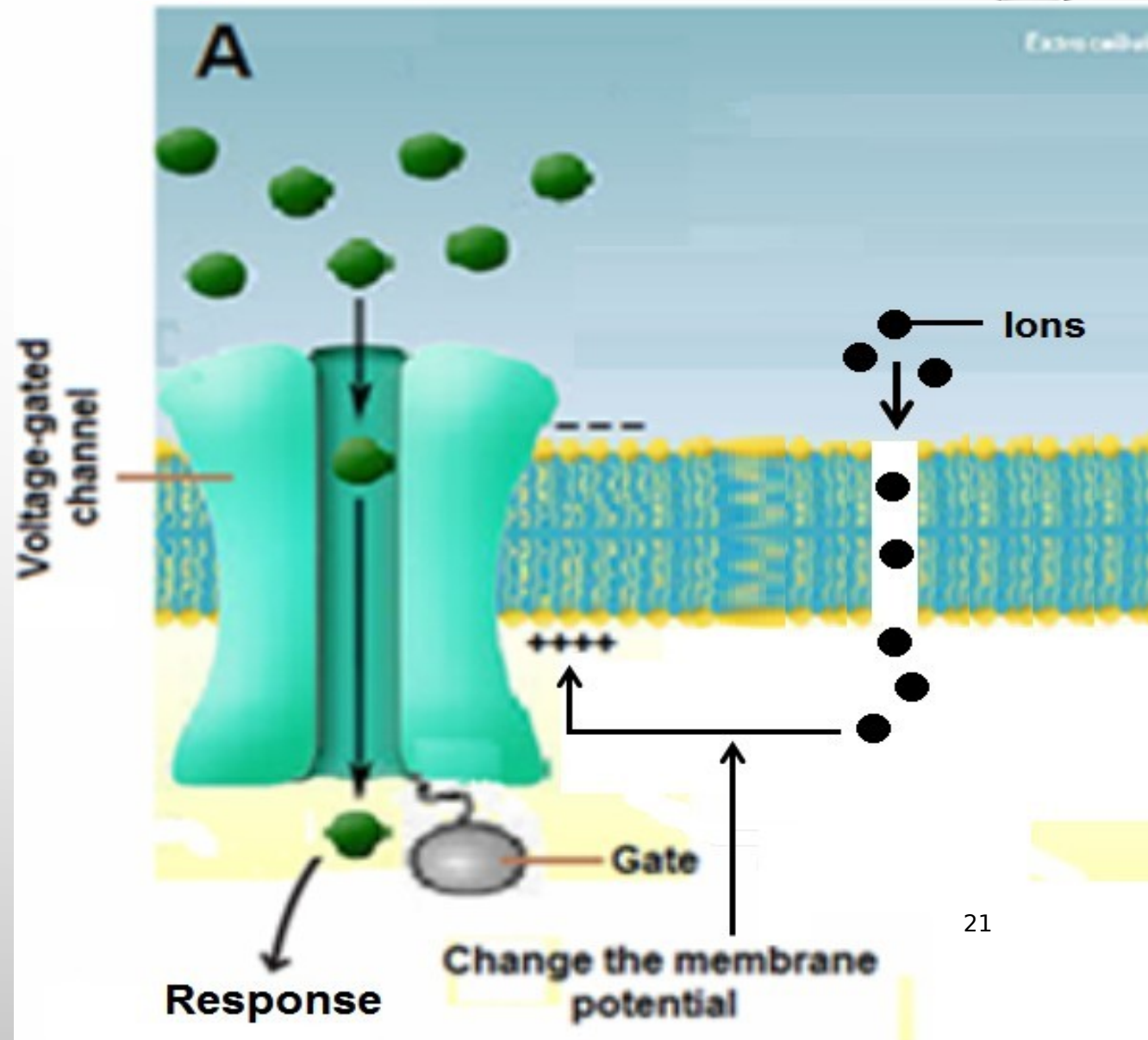


Electrical, physical or
chemical signal input signal



[A] Electrical communication:

- flow of ions → change in cell mem. pot. → activation of voltage gated ion channels → influx or efflux of ions → change in chemical composition of the cell → response



B. Chemical communication:

1. Endocrine pathway, ACTH from ant. pituitary → cortisol

2. Paracrine Pathway, somatostatin → decrease secretin

3. Autocrine pathway

4. Contact-dependent pathway

5. Intracrine pathway, steroids

➤ Chemical communication involves binding of 1st messenger (ligand) to the receptors. The result of the binding is one of the following:

- [1] Open or close of Ion channel-linked receptors
- [2] Activation or inhibition of G- proteins
- [3] Activation or inhibition of receptor-linked guanylyl cyclase (catalysis-linked receptors)
- [4] Activation or inhibition of receptor-linked tyrosine kinase (catalysis-linked receptors)
- [5] Gene modification

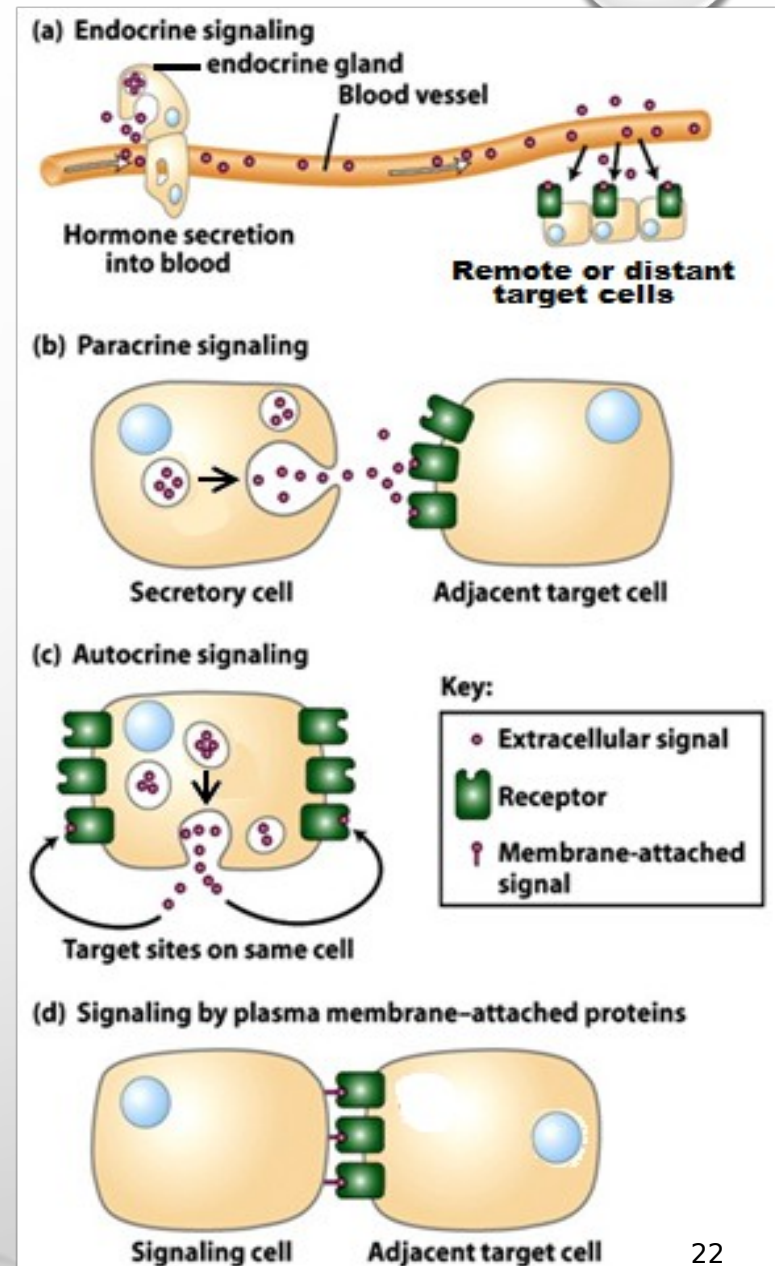
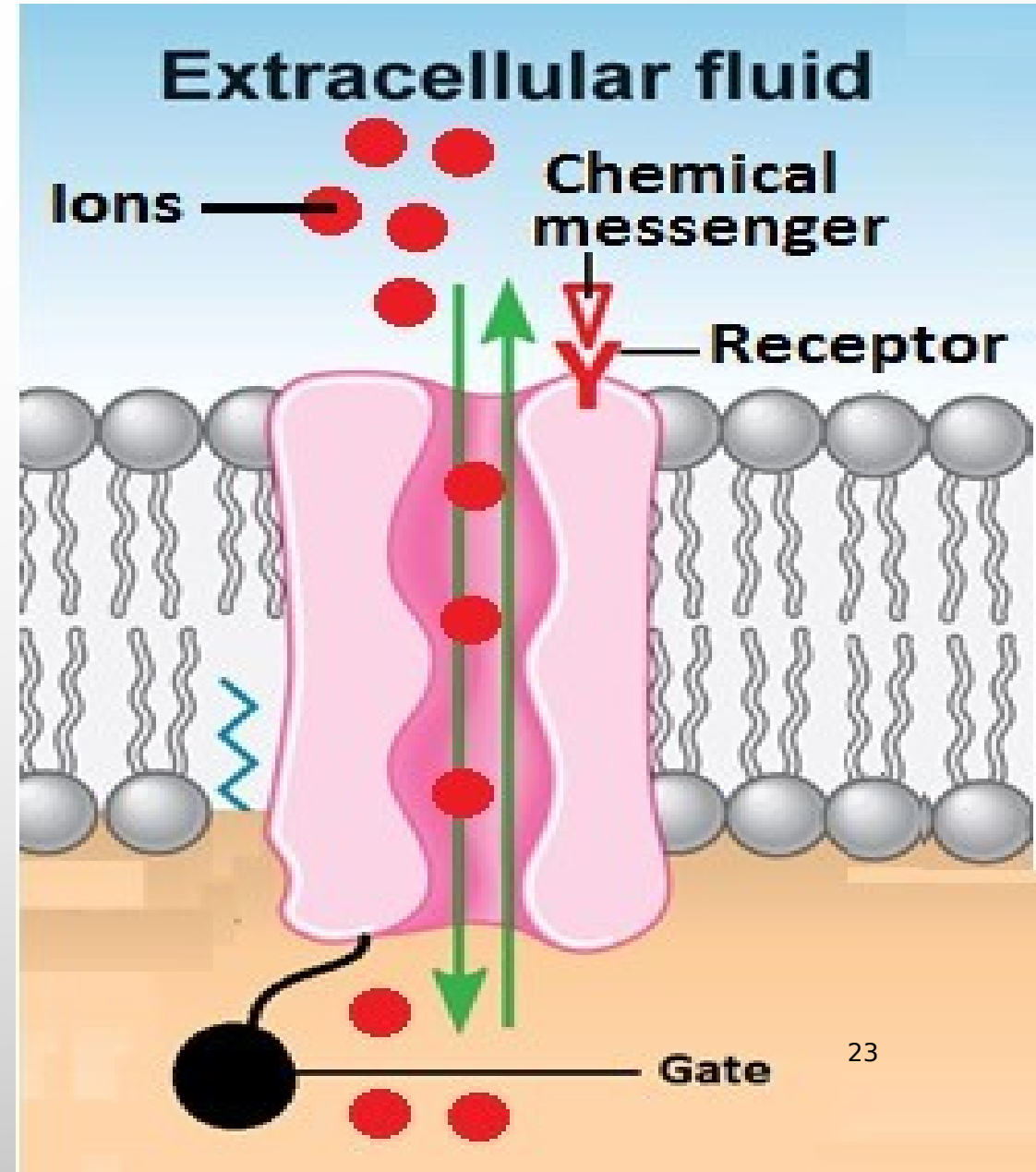


Figure 1.8: Intercellular chemical communication types.

[1] Open or close of Ion channel-linked receptors

- Conformational or structural change in the ion-gated channel protein attached to the receptor
- lead to open or close the gate → →
- increase or decrease in the diffusion of ions



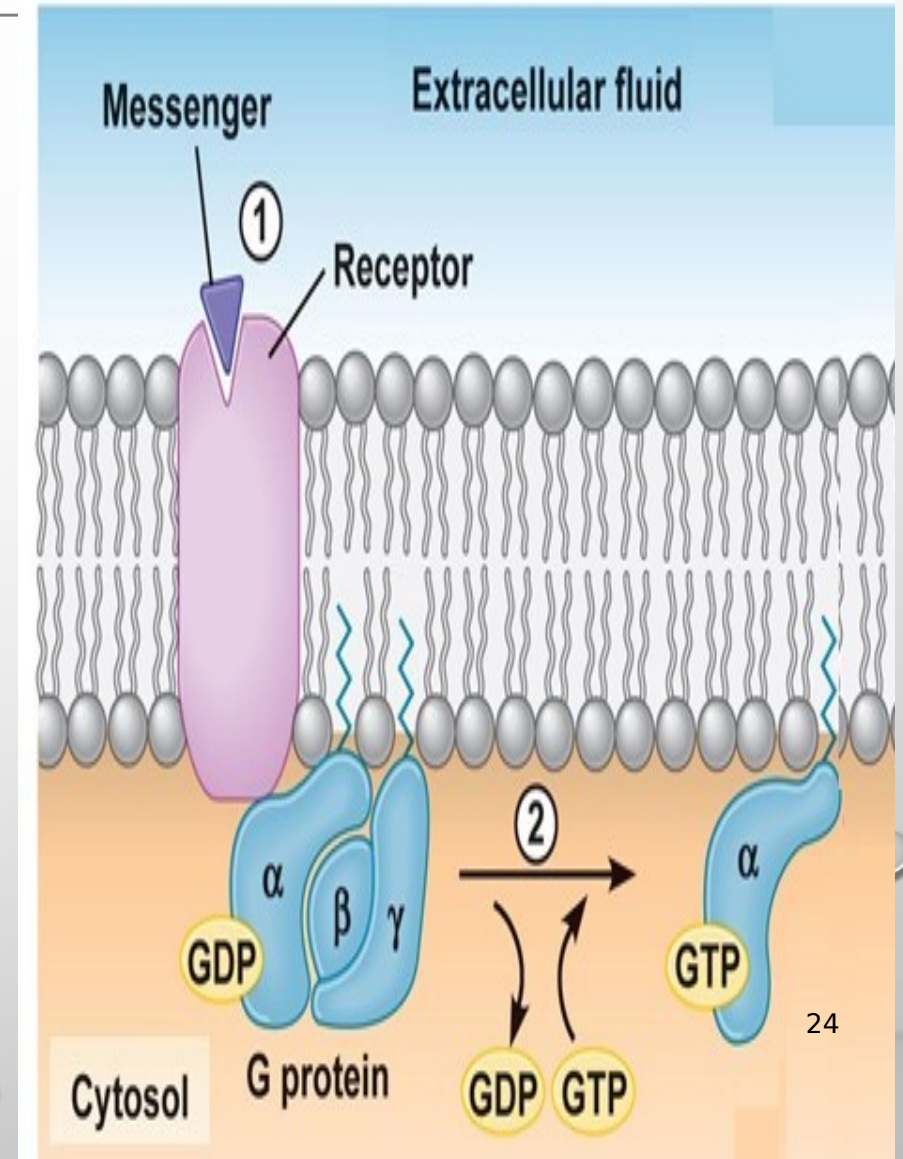
[2] Activation or inhibition of G- proteins

The α subunit can then migrate laterally in the plasma membrane to modulate the activity of:

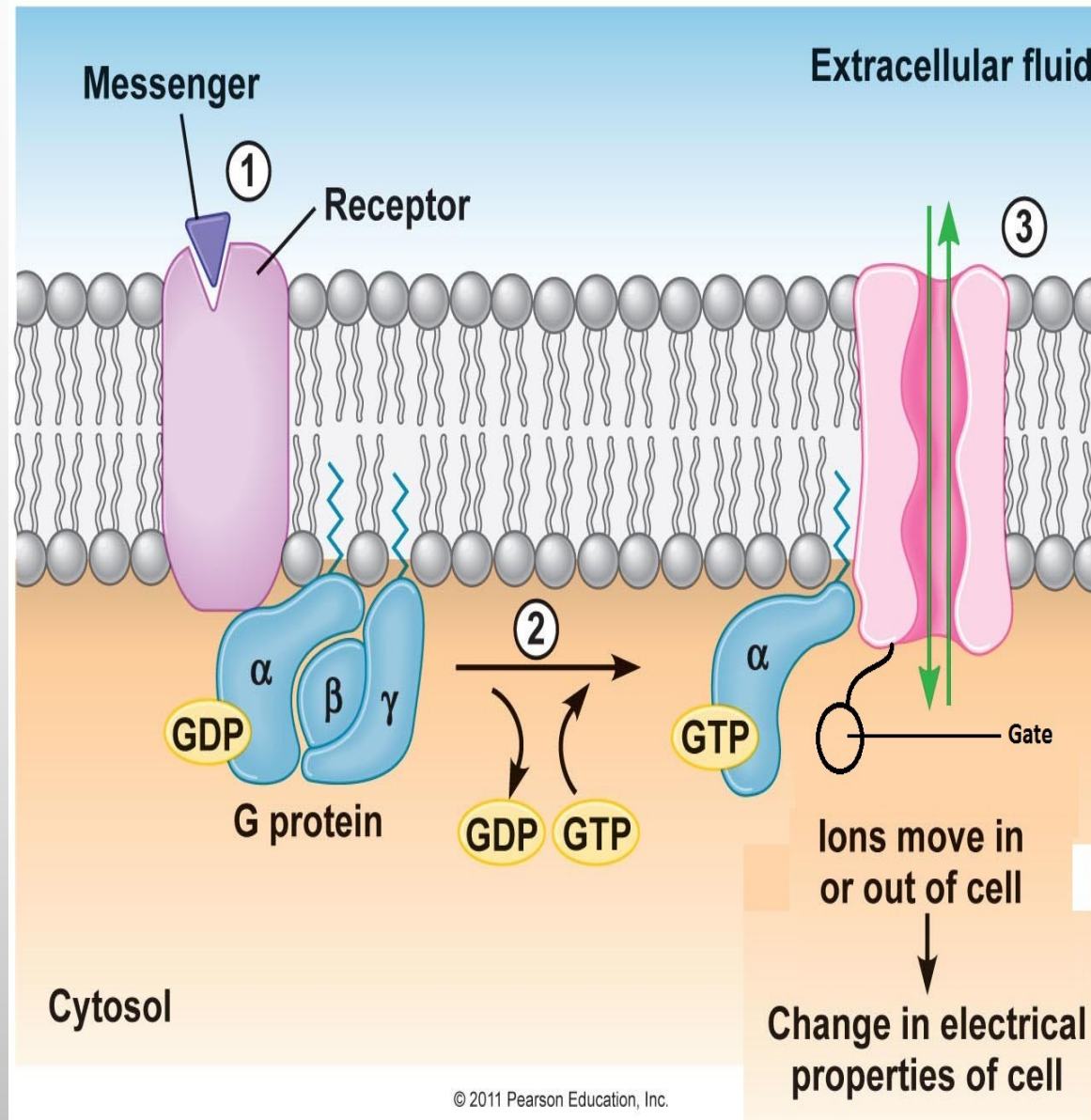
⊙ G protein-linked ion channels

⊙ G protein-linked activation or inhibition of adenylate cyclases

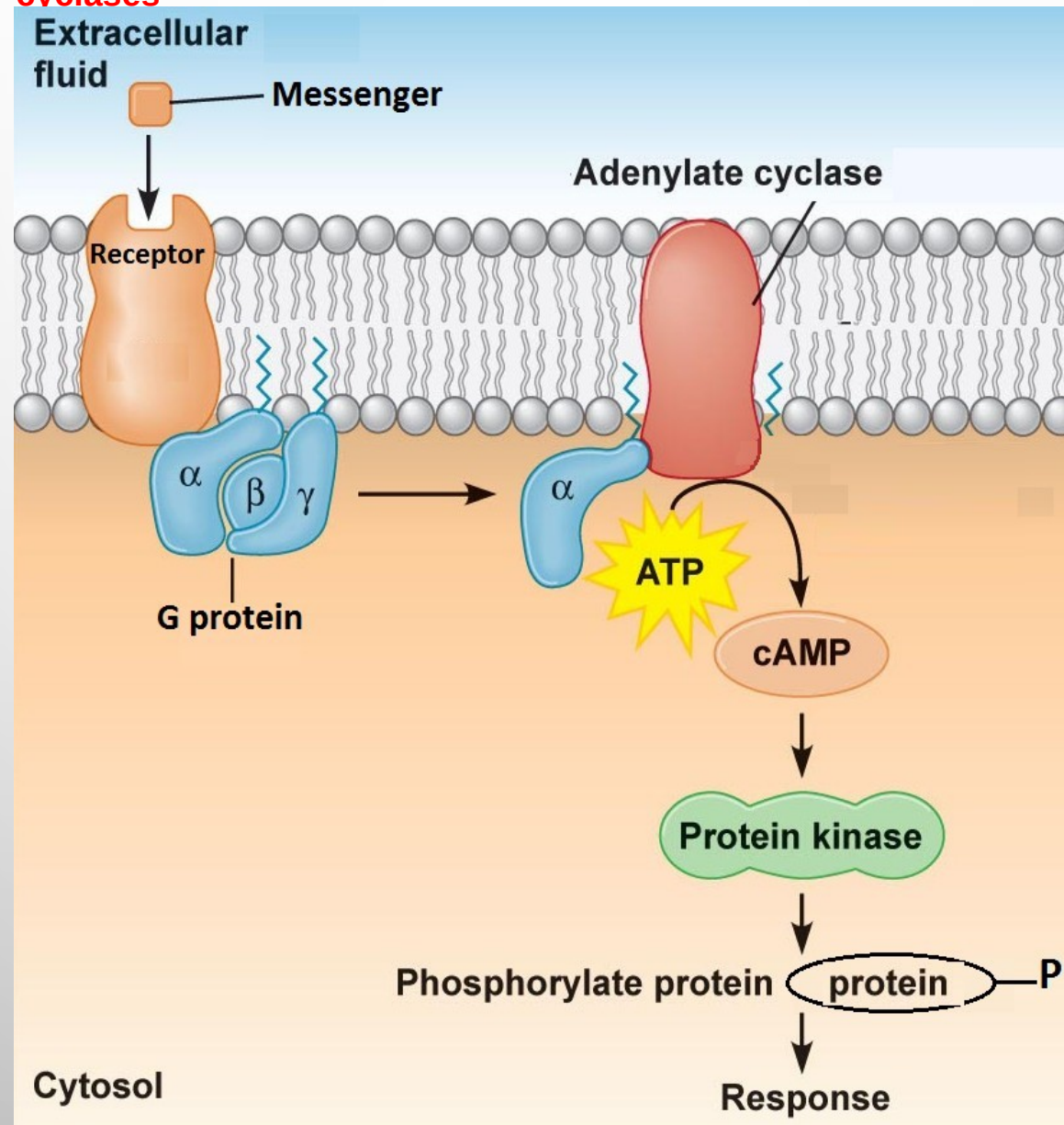
⊙ G protein-linked activation or inhibition of phospholipases C



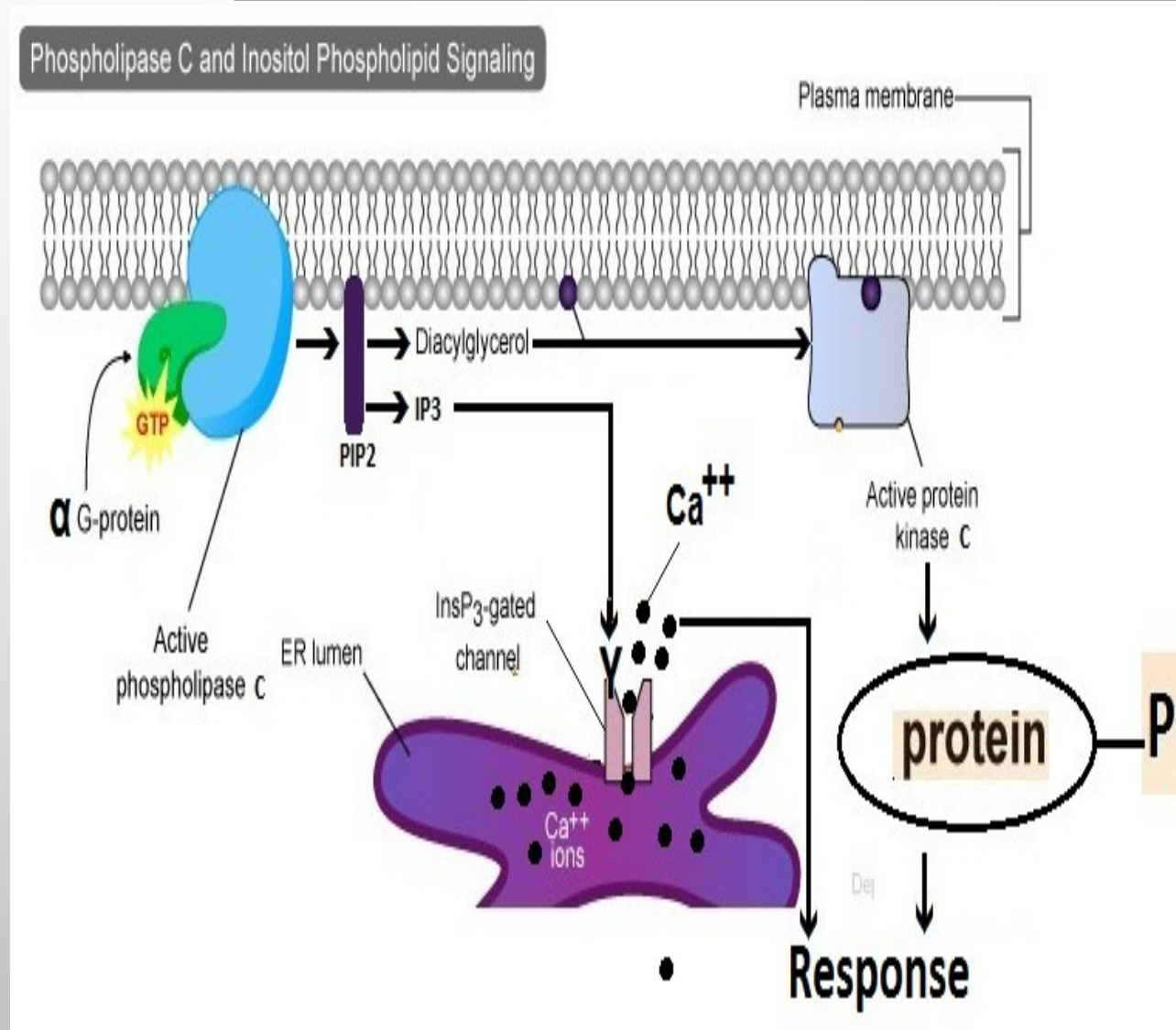
[A] G protein-linked ion channels



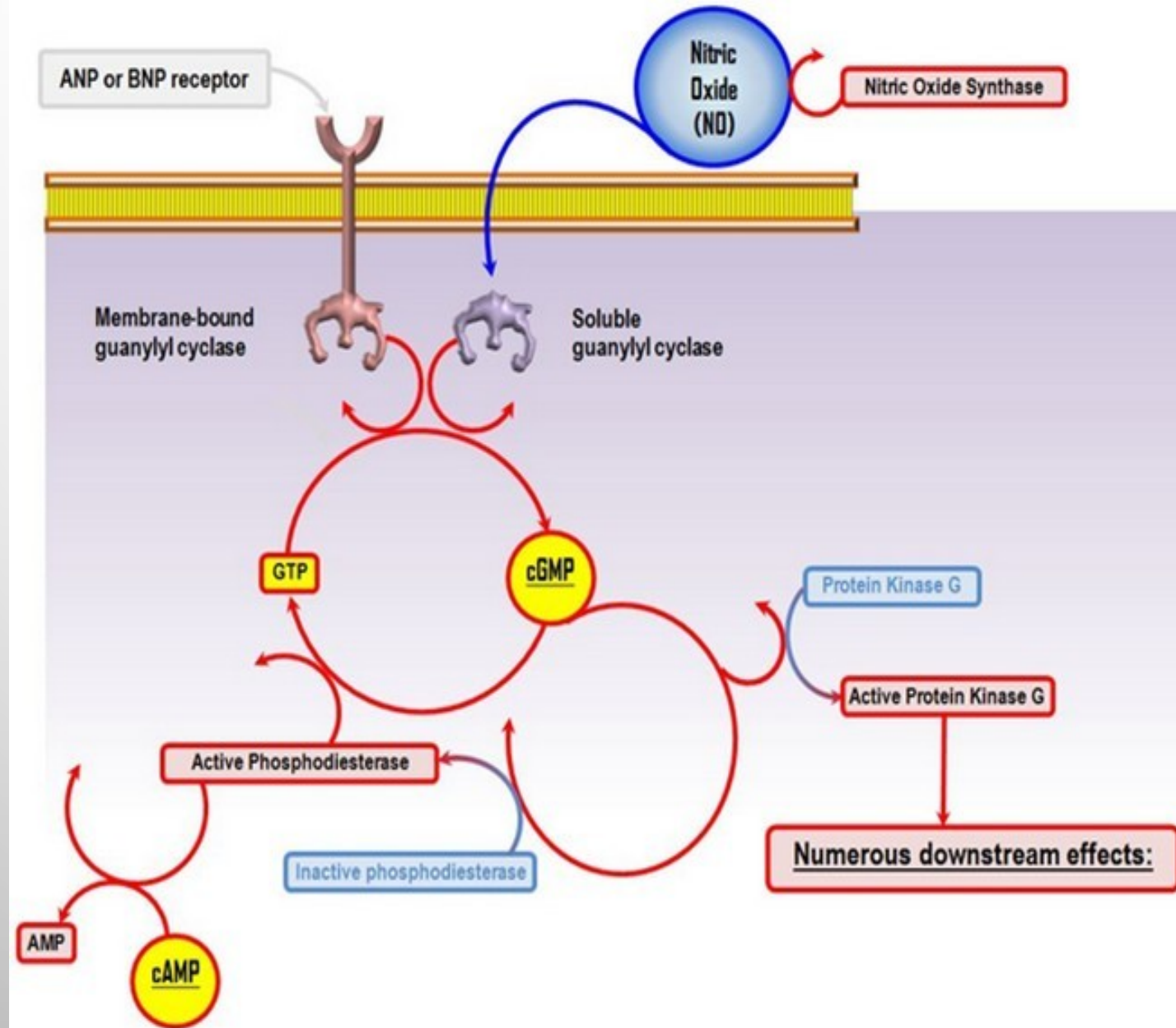
[B] Activation or inhibition of G protein-linked adenylate cyclases



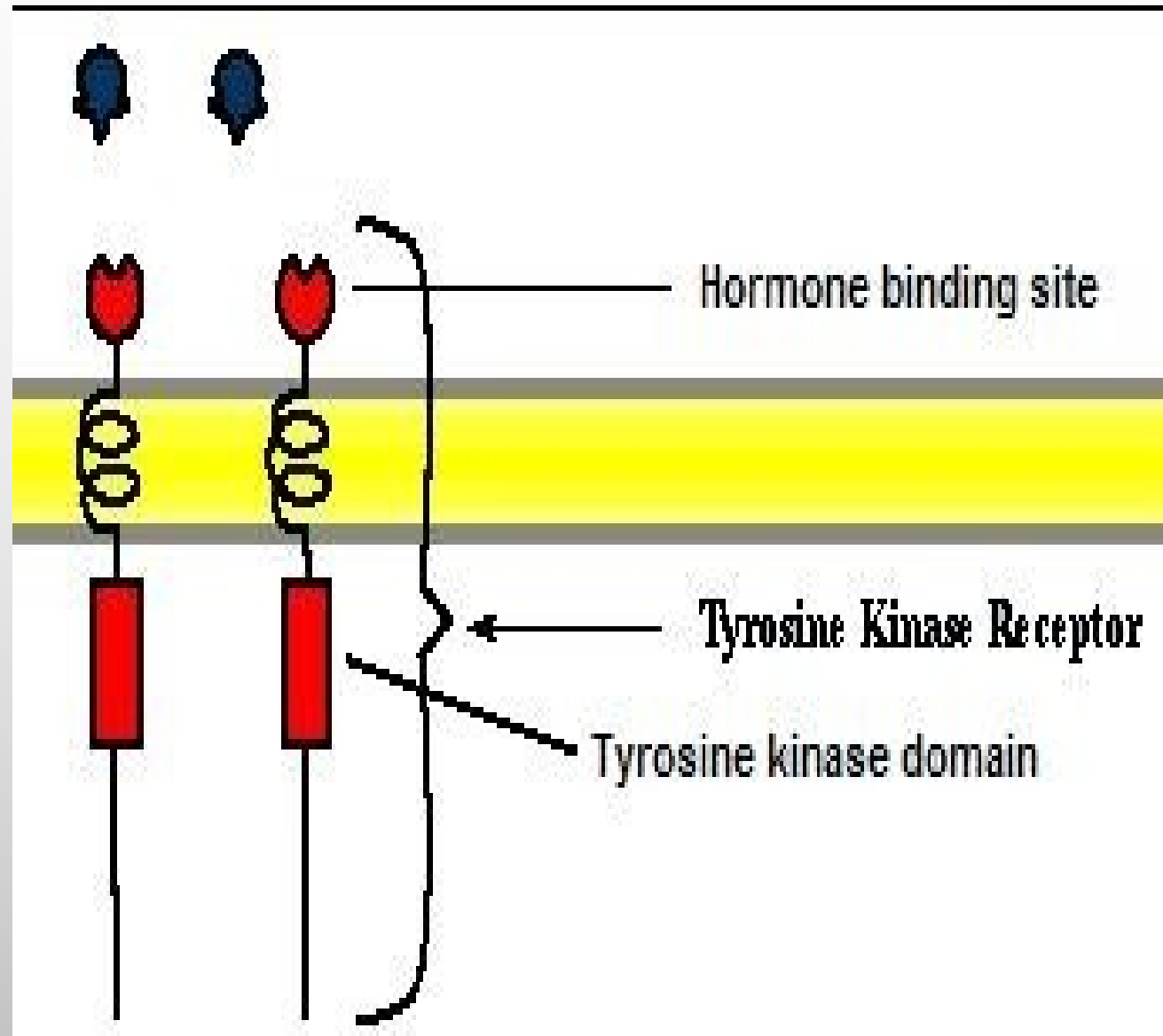
[C] Activation or inhibition of G protein-linked phospholipases C



[3] Activation or inhibition of receptor-linked guanylyl cyclase (catalysis- linked receptors)

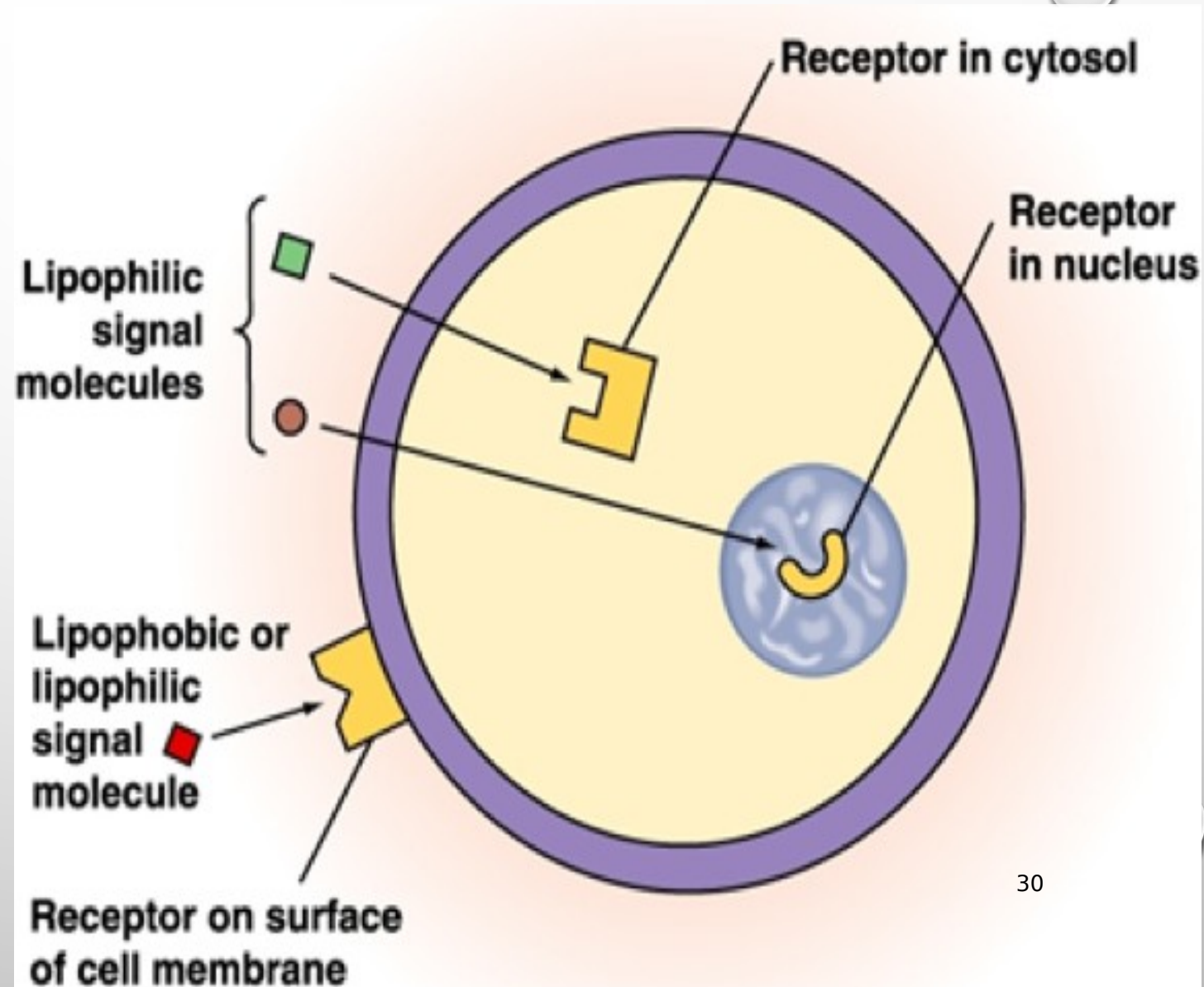


[4] Activation or inhibition of receptor-linked tyrosine kinase (catalysis-linked receptors)

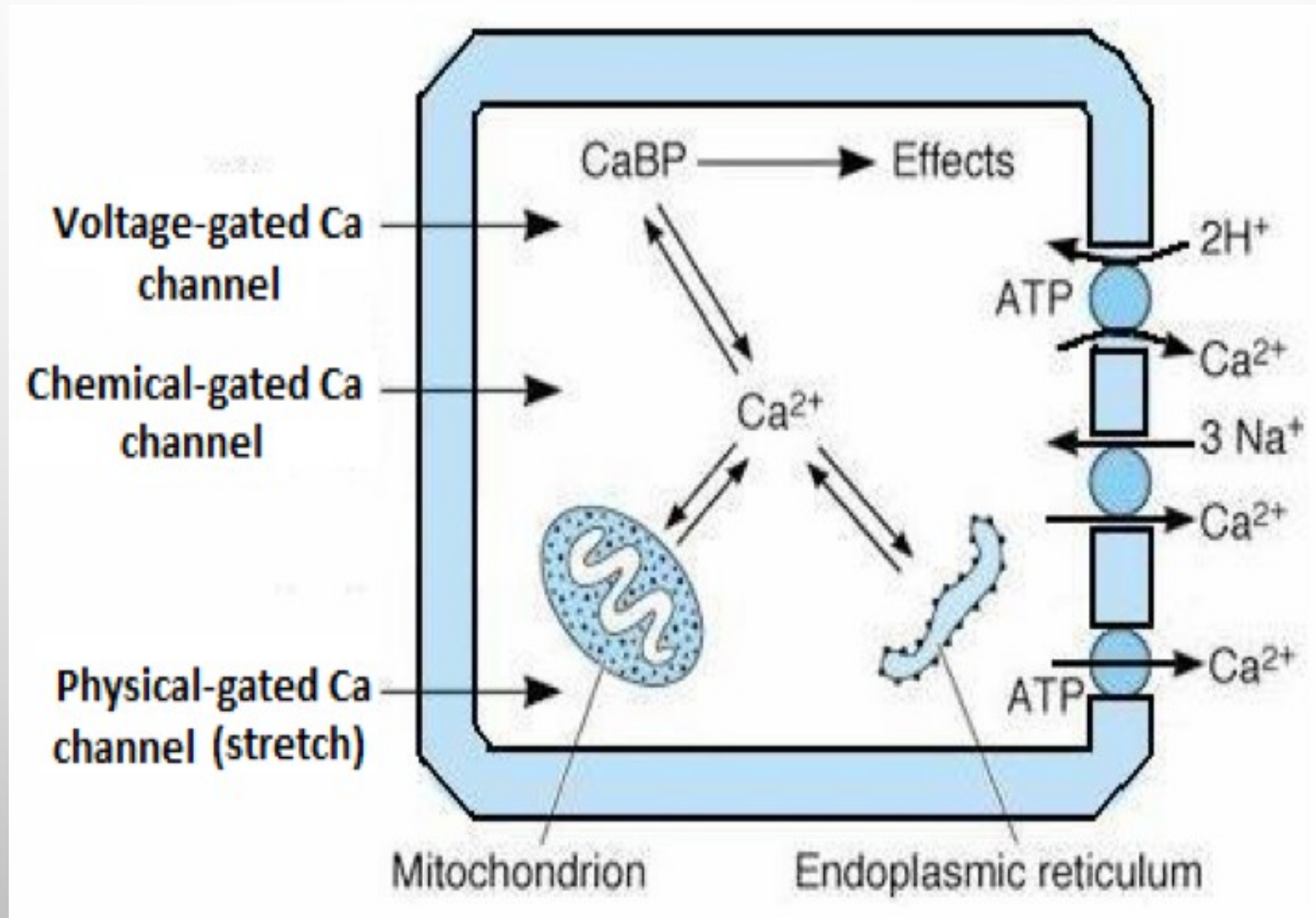


[5] Gene modification

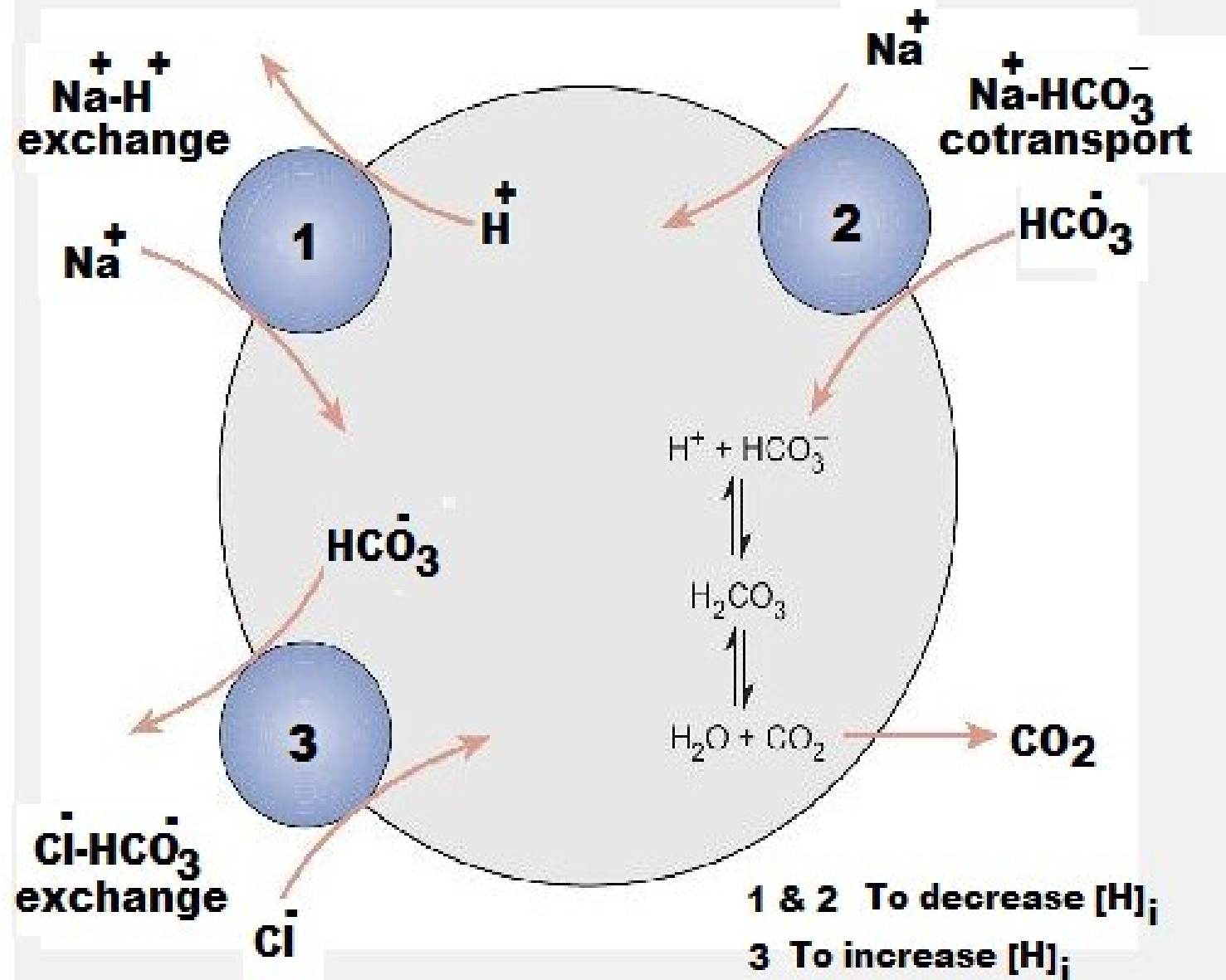
- chemical messenger bind to specific receptor → conformation of the receptor → gene → more increased the formation of specific protein molecule with enz. activity



REGULATION OF THE INTRACELLULAR Ca^{2+}



REGULATION OF INTRACELLULAR [H]:



CELL VOLUME REGULATION

IT DEPENDS ON THE TOTAL AMOUNT OF INTRACELLULAR SOLUTE (Na^+ AND K^+ IONIC GRADIENTS) GENERATED BY THE Na^+/K^+ PUMP

- **Following cell shrinkage** □ intracellular solute concentration increasing mechanism is activated
- □ the synthesis of osmotically active molecules (eg, sorbitol or taurine)
- □ the transport of ions inside the cell through the Na^+/H^+ exchanger or the $\text{Na}^+/\text{H}^+/\text{Cl}^-$ cotransporter
- **Following cell swelling** □ transport mechanisms that extrude solutes out of the cell (eg, K^+ or Cl^- channels or the K^+/Cl^- cotransporter) will be activated.

Cell Apoptosis = Cell suicide □ **Genetically
determined**

Cell Necrosis = Cell murder □ **External cause**